# Tokenized Deposits — Architecture & Technical Flow

## Overview

Tokenized Deposits is a system that represents fiat currency deposits as ERC-20 tokens on a blockchain. When a user deposits USD, tokens are minted to their wallet. When they withdraw, tokens are burned. The token balance is always a 1:1 mirror of their fiat position.

### System Architecture

Syntax error in text

mermaid version 11.14.0

## Directory Structure

```
tokenized-deposits/
├── backend/                    # Python FastAPI service
│   ├── main.py                 # App entry point, lifespan, middleware
│   ├── routers/
│   │   ├── clients.py          # All client-facing endpoints
│   │   └── admin.py            # Operator-only endpoints
│   ├── services/
│   │   ├── kyc.py              # KYC verification stub
│   │   ├── wallet.py           # Ethereum address generation
│   │   └── event_listener.py   # Background on-chain event poller
│   └── tests/                  # pytest test suites
│
├── blockchain/                 # Hardhat project
│   ├── contracts/
│   │   └── DepositToken.sol    # The ERC-20 token contract
│   ├── scripts/
│   │   └── deploy.ts           # Deploy script (writes to Firestore)
│   └── test/
```

```
│       └── DepositToken.test.ts
│
├── frontend/                   # Flutter app
│   ├── lib/
│   │   ├── main.dart           # App entry, global providers, session restore
│   │   ├── config/app_config.dart
│   │   ├── models/             # Plain Dart data classes
│   │   ├── providers/          # Riverpod state notifiers
│   │   ├── screens/            # UI screens
│   │   └── services/           # ApiClient, SessionService
│   └── test/                   # Flutter widget tests
│
├── firestore.rules             # Deny all client-side access (backend-only via Admir
└── firestore.indexes.json      # Composite index definitions
```

---

## The Smart Contract — `DepositToken.sol`

**File:** `blockchain/contracts/DepositToken.sol`

Each `(asset_type, network)` pair gets its own deployed instance of this contract. E.g. USD on hardhat is one contract, EUR on Sepolia is another.

```
contract DepositToken is ERC20, Ownable, Pausable {
string public assetType;      // e.g. "USD"

string public networkLabel;   // e.g. "hardhat"

mapping(address => bool) private _approved;  // KYC allowlist
```

### Contract Inheritance

Syntax error in text

mermaid version 11.14.0

**Key design decisions:**

| Feature | How it works |
|---|---|
| Access control | `onlyOwner` — only the operator wallet (backend) can mint/burn |
| KYC allowlist | `registerWallet(address)` adds a wallet; mint/burn revert with `WalletNotApproved` if not in the list |
| Pause circuit breaker | `pause()` / `unpause()` block all mint/burn operations; useful for incident response |
| Events | `Mint(address indexed recipient, uint256 amount)` and `Burn(address indexed source, uint256 amount)` are emitted and used by the event listener |

**Deployment:** `blockchain/scripts/deploy.ts` deploys a contract, saves the address to a local JSON file, and writes a record to the Firestore `token_registry` collection so the backend can discover it at startup.

---

## Firestore Collections

All Firestore access is via the Firebase Admin SDK (backend only). Client-side access is denied in `firestore.rules`.

## Collection Schema

Syntax error in text

mermaid version 11.14.0

`clients/{client_id}`

```
{
"id": "uuid",

"first_name": "Alice",

"last_name": "Smith",

"date_of_birth": "1990-01-15",

"national_id": "AB1234",

"kyc_status": "approved",

"kyc_failure_reason": null,

"wallet": {

"hardhat": "0xABC...",

"sepolia": "0xDEF..."

}

}
```

`transactions/{tx_id}`

```
{
"id": "uuid-or-tx-hash",

"client_id": "uuid",

"type": "deposit",

"amount": 100,

"asset_type": "USD",

"network": "hardhat",

"status": "confirmed",

"on_chain_tx_hash": "0x...",

"created_at": "2026-04-05T10:00:00Z"

}
```

`token_registry/{asset_type}_{network}`

```
{
"contract_address": "0x...",

"asset_type": "USD",

"network": "hardhat",

"deployed_at": "2026-04-05T09:00:00Z",

"deployer_address": "0x..."

}
```

`system/event_listener`

```
{
"last_processed_block_hardhat": 42,

"last_processed_block_sepolia": 17000000

}
```

Used as a cursor so the event listener knows which blocks have already been processed.

---

## Backend Startup Sequence

**File:** `backend/main.py`

Syntax error in text

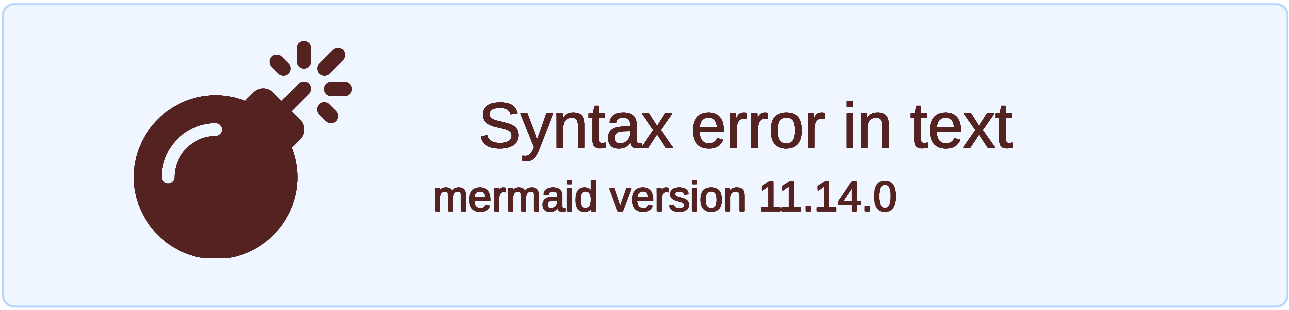
mermaid version 11.14.0

The `token_registry` dict is kept in `app.state` and shared across all requests. The event listener refreshes it on every poll cycle so newly deployed contracts are picked up without a restart.

---

## User Flows

### 1. App Launch & Session Restore




---

### 2. KYC + Wallet Creation

**KYC failure path:**

---

## 3. Deposit (Fiat → Tokens)

---

## 4. Withdrawal (Tokens → Fiat)

---

## 5. Event Listener (Background Sync)

**File:** `backend/services/event_listener.py`

**Topic computation:**

```
MINT_TOPIC = Web3.keccak(text="Mint(address,uint256)").hex()
BURN_TOPIC  = Web3.keccak(text="Burn(address,uint256)").hex()
```

These match the `event Mint(address indexed recipient, uint256 amount)` signature in the Solidity contract.

---

### 6. Admin — Reconciliation

**Endpoint:** `GET /admin/reconcile` (requires `X-API-Key` header)

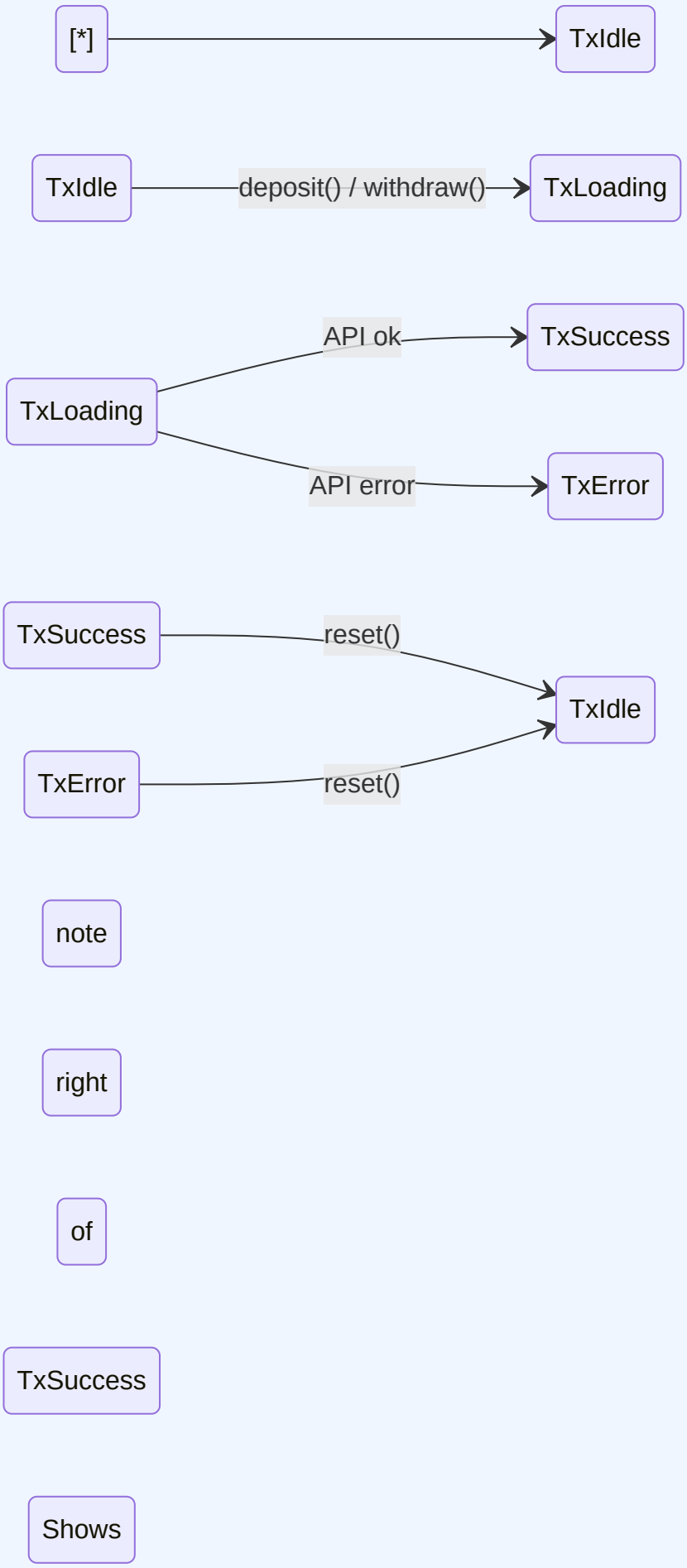
---

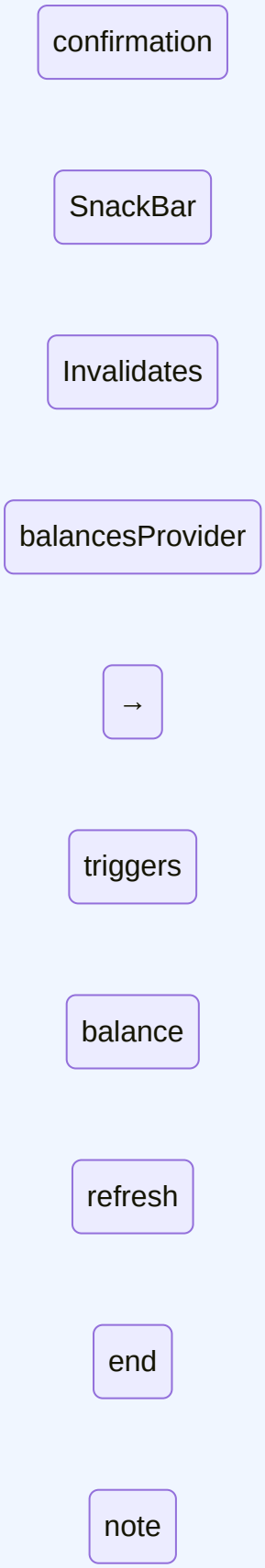
## Frontend State Management

**Library:** Riverpod 2.x (`flutter_riverpod`)

### Provider Dependency Graph

## Sealed State Classes


[*]
TxIdle
TxIdle
deposit() / withdraw()
TxLoading
API ok
TxSuccess
TxLoading
API error
TxError
TxSuccess
reset()
TxIdle
TxError
reset()
note
right
of
TxSuccess
Shows




## Session persistence (`lib/services/session_service.dart`)

Uses `shared_preferences` to persist the `clientId` and wallet JSON to device/browser storage. On startup, `main()` reads these before `runApp()` and seeds the providers via `ProviderScope`

overrides, so the user's session survives page refreshes and app restarts.

```
// main() startup sequence
WidgetsFlutterBinding.ensureInitialized();

final savedClientId = await SessionService.loadClientId();

final savedWallet   = await SessionService.loadWallet();

runApp(ProviderScope(

overrides: [

currentClientIdProvider.overrideWith((ref) => savedClientId),

currentWalletProvider.overrideWith((ref) => savedWallet),

],

child: const TokenizedDepositsApp(),

));
```

---

## API Reference

| Method | Path | Auth | Description |
|---|---|---|---|
| POST | `/clients` | — | KYC verification + client creation |
| POST | `/clients/{id}/wallet` | — | Generate chain addresses + on-chain registration |
| POST | `/clients/{id}/deposit` | — | Mint tokens for a fiat deposit |
| POST | `/clients/{id}/withdraw` | — | Burn tokens for a fiat withdrawal |
| GET | `/clients/{id}/balance` | — | On-chain balance for one `(asset_type, network)` |
| GET | `/clients/{id}/balances` | — | On-chain balances for all token registry pairs |
| GET | `/clients/{id}/transactions` | — | Firestore transaction history |
| POST | `/admin/pause` | `X-API-Key` | Pause a DepositToken contract |
| POST | `/admin/unpause` | `X-API-Key` | Unpause a DepositToken contract |
| GET | `/admin/reconcile` | `X-API-Key` | Compare on-chain vs. Firestore balances |
| GET | `/health` | — | Liveness check |

## Running Locally

**Prerequisites:** Python 3.11+, Node 18+, Flutter 3.x, Firebase project with Firestore.

```
# 1. Start the local blockchain
cd blockchain

npm install

npm run node                          # keeps running in terminal 1
```

## 2. Deploy a token contract (terminal 2)

```
npx hardhat run scripts/deploy.ts --network localhost \

-- --asset-type USD --network-label hardhat
```

## 3. Start the backend (terminal 3)

```
cd backend

python -m venv venv && source venv/bin/activate

pip install -r requirements.txt

cp secrets/firebase-credentials.json ...   # populate from Firebase Console

uvicorn main:app --reload
```

## 4. Start the Flutter app (terminal 4)

```
cd frontend
```

```
flutter run -d chrome \

--dart-define=BASE_API_URL=http://localhost:8000
```

**Environment variables (backend `.env`):**

| Variable | Purpose |
|---|---|
| `OPERATOR_PRIVATE_KEY` | Hardhat/Sepolia account that owns the contracts |
| `HARDHAT_RPC_URL` | Defaults to `http://127.0.0.1:8545` |
| `SEPOLIA_RPC_URL` | Alchemy/Infura URL for Sepolia testnet |
| `ADMIN_API_KEY` | Secret for `/admin/*` endpoints |
| `GOOGLE_APPLICATION_CREDENTIALS` | Path to Firebase service account JSON |

---

## Key Design Patterns

**Dual-write consistency:** Every deposit/withdrawal creates a Firestore record *and* submits an on-chain transaction. The event listener provides a safety net by independently reading on-chain events and creating Firestore records if they are missing, keeping the two stores in sync.

**Operator key model:** The backend operator wallet is the sole `owner` of all DepositToken contracts. Clients never hold private keys that can mint or burn — they only hold addresses where tokens are credited. This means the backend controls all token operations.

**Idempotency:** The event listener keys transaction records on `on_chain_tx_hash`, so replaying events (e.g. after a restart) never creates duplicates.

**Test overrides:** `apiClientProvider` is a plain `Provider` that tests replace via `ProviderScope(overrides: [...])`. Backend tests patch `main.run_event_listener` with `AsyncMock` to prevent the background task from running during test setup.